Institute of Technology of Cambodia
Department of Mechanical and Industry

Subject: **Finite Element Method (Hook)**

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Group: **I4-GIM-Mechanical**

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Hook

Exercise [20pt]

A hook and its hanging system are shown in the figure below. Detailed dimensioı can be obtained from image software such as Image-J or Microsoft Paint. This hook is maα of steel with E = 200 GPa and = 0.3. Its plastic property can be used from the lectu exercise “Book1.xls”.

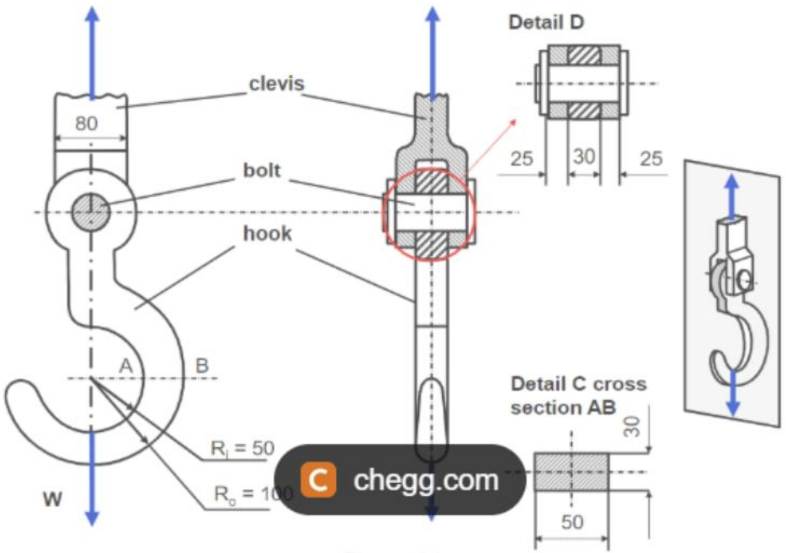


Figure Q1

convergence for example: adding an arc as analytical rigid body in order to apply the load W instead of apply a Concentrated Force.

2) If there is a convergence, please use the mesh size from which convergence occurs to determine the maximum load (W_{max}) the hook can withstand without plastic deformation. If not, please use the finest mesh size your computer can run.

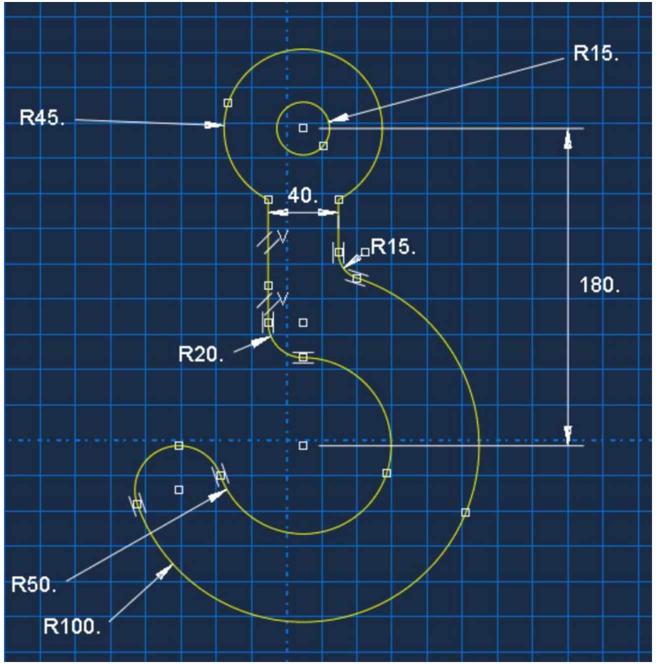
Presentation: *please prepare all your Abqus results into a presentation file. You will have 10 mn each for a presentation so the maximum number of slides should not be more than 15 pages. Please make the slides following below guidance:*

- Modelling objectives
- How you model your Abaqus (preferably after you finish applying load and boundary conditions)
- How you do the mesh
- What are your results (answer to the above two questions) as function of 3 mesh sizes.
- Discussion on your results
- Finally conclude your work.

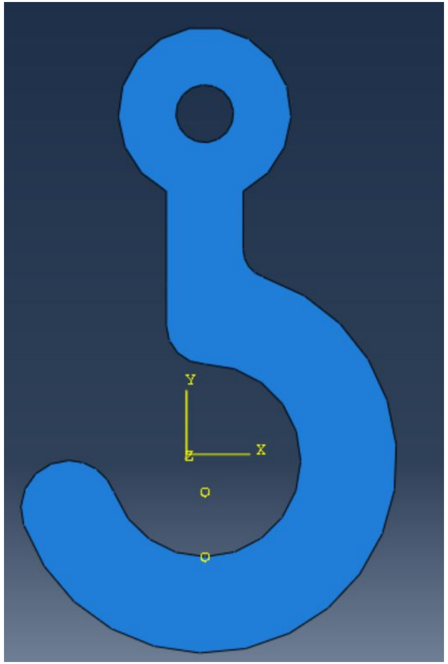
Contents

Geometry of hook

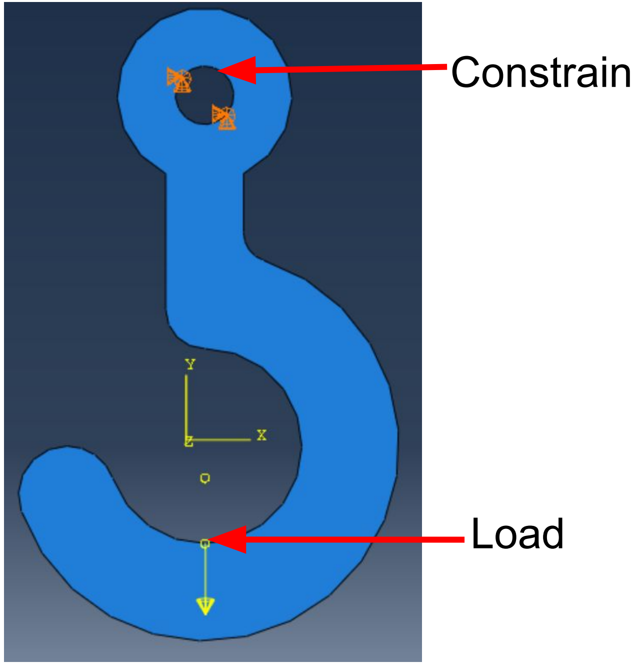
Thickness = 30 mm



Before



After



Properties of ASTM-A16

Elastic Core Properties

Young's Modulus (E)**200 GPa**

Poisson's Ratio (v)**0.3**

Yield Strength (σy) **250 MPa**

Plastic Core Properties

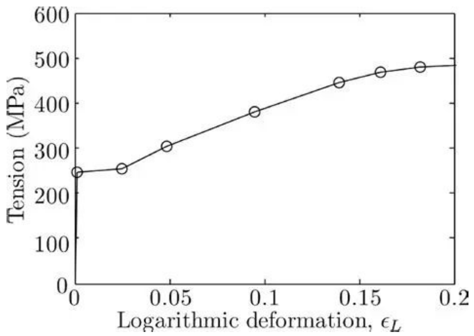
True Stress : $\sigma T = \sigma \ (1 + \varepsilon)$

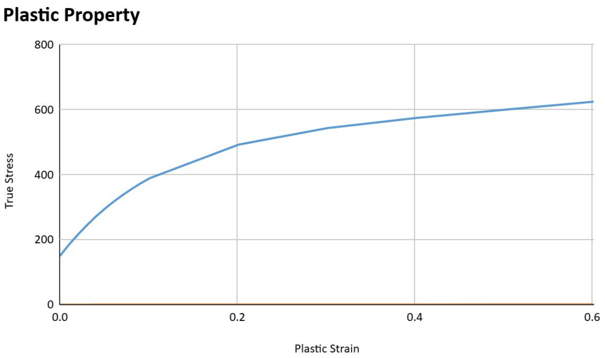
True Strain : $\varepsilon T = \ln(1 + \varepsilon)$

Plastic Strain: $\varepsilon = \varepsilon\_n - \varepsilon\_1$

True Stress	Plastic Strain
150.0000417	0
184.4384466	0.01017213704
215.9392217	0.02032946018
244.765057	0.03047324286
271.1550407	0.04060464756
295.3268264	0.05072474924
317.4785851	0.06083450113
337.7908006	0.07093481428
356.4278597	0.08102649435
373.539523	0.09111028252
389.2622437	0.1011868579
492.7103955	0.2016694033
543.5099521	0.3018655867
574.8712485	0.4019453572
600.1934356	0.5019777819
624.9388371	0.6019909689

σy = 250 MPa





Simulation Constraints

Incremental Solver Step

Specifies step-size scaling limits assigned for incremental stress stabilization iterations.

Step = 0.25

Surface Interaction

Analytical rigid body constraints configured as tie interactions across model surfaces.

Type: Tie

Boundary Constraints

Rigid translation pin connections restricting key axial motion degrees of freedom.

Type: Pin

Gravity & Force Load

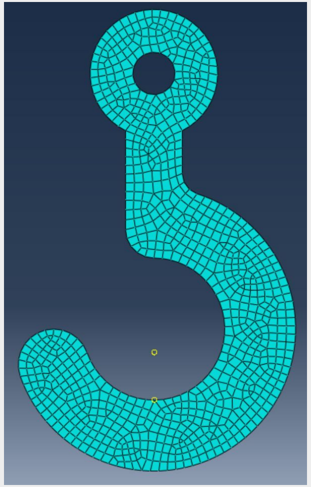
Concentrated load converted to Newton vector metrics derived from structural mass variables.

F = mg = 1000 *9.81

9,810 N

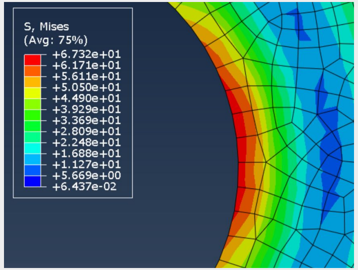
Global Mesh Seed

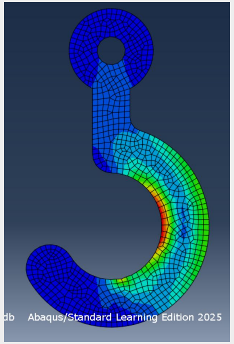
Global Mesh seed



- 5.5 mm = 1013 elements
- 5.7 mm = 936 elements
- 5.8 mm = 913 elements
- 6 mm = 857 elements
- 8 mm = 486 elements
- 10 mm = 316 elements
- 15 mm = 155 elements

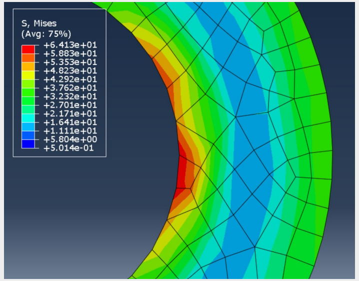
Global Mesh seed 6 mm

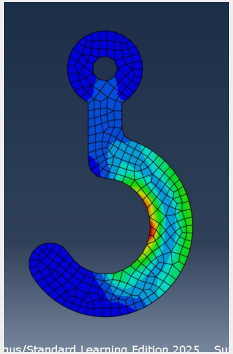




Max. von Mises Stress = 67.8848 MPa

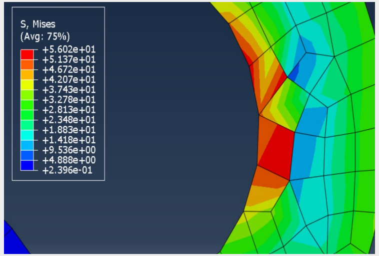
Global Mesh seed 10 mm

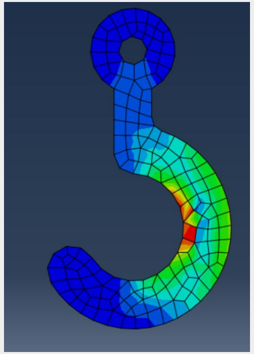




Max. von mise stress = 67.2927 MPa

Global Mesh seed 15 mm

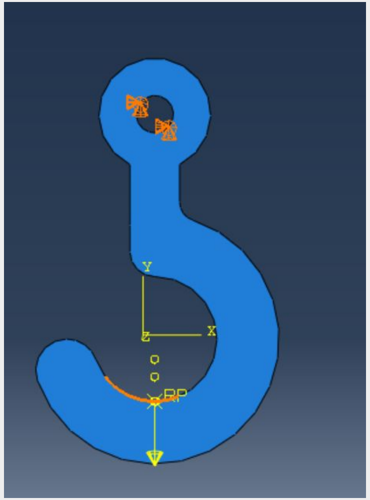




Max. von mise stress = 59.0191 MPa

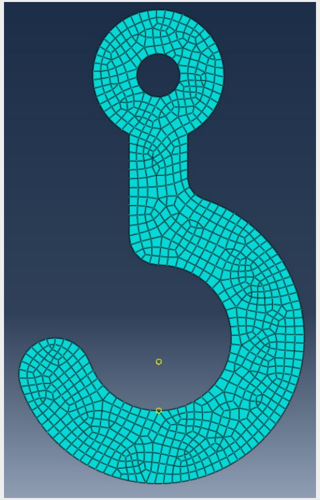
Applying Analytical Rigid Body

Geometry Setup



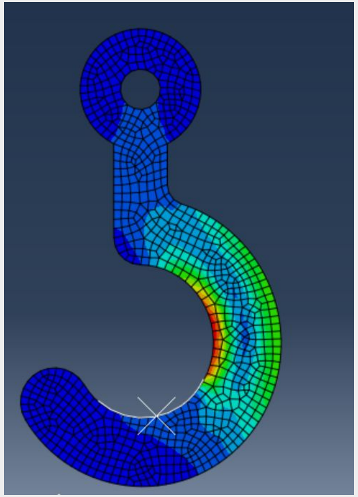
With the analytical rigid body.

Mesh Generation

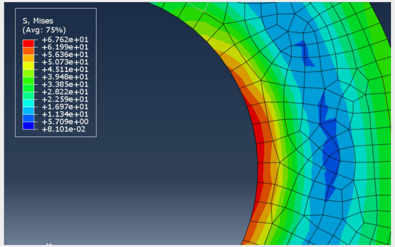


Mesh Seed 6 mm (857 elements)

Von Mises Stress Results



Distributed contact pressure avoids artificial point-load stress singularities.

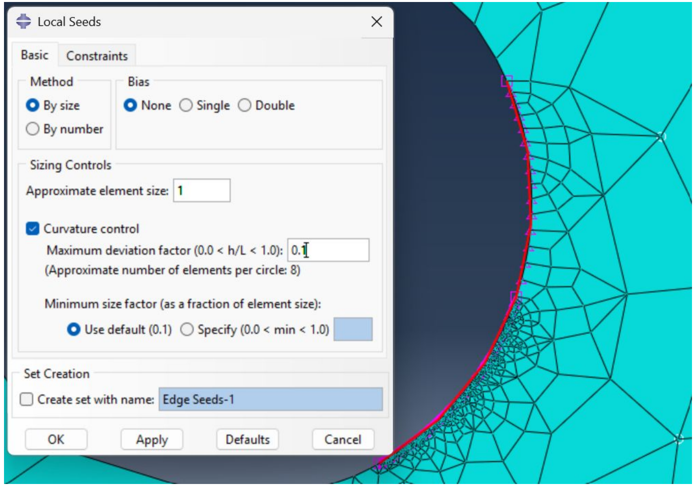


Max. von Mises Stress = 67.1814 MPa

Localized Mesh Refinement Strategy

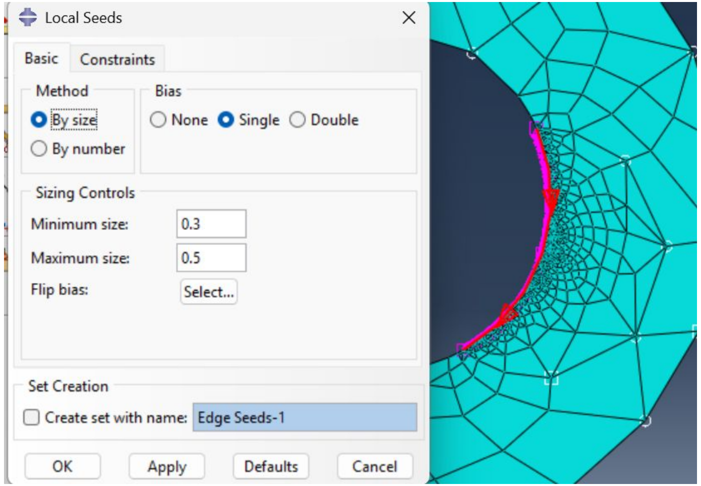
Alternative Seeding Formulations

Method A: Curvature-Based



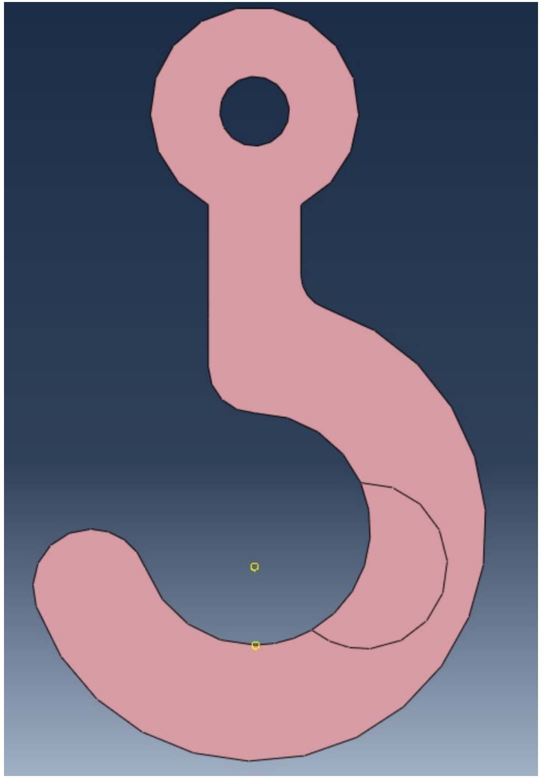
Geometric curvature limits elements to 0.5 mm at throat.

Method B: Biased Graded Seeding

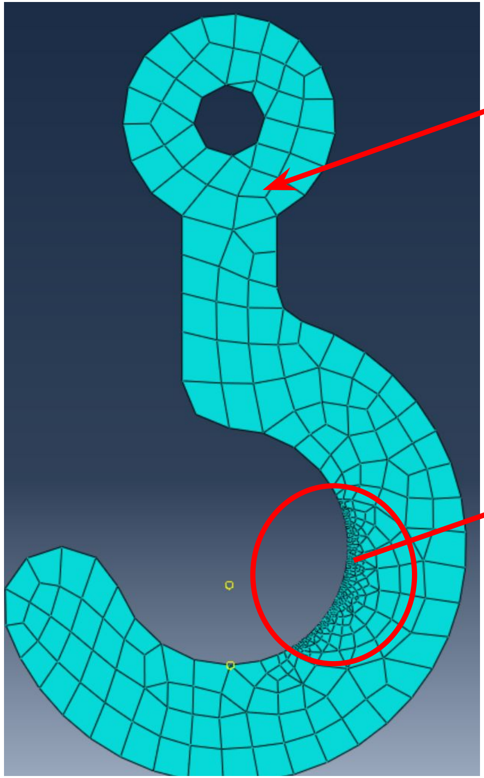


Biased transition spans from 0.075 mm to 0.35 mm nodes.

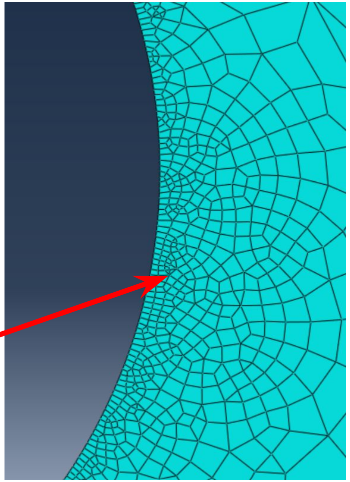
Localized Mesh Refinement Strategy



Geometric Partitioning of Inner Throat Region



Global Mesh
seed:15mm

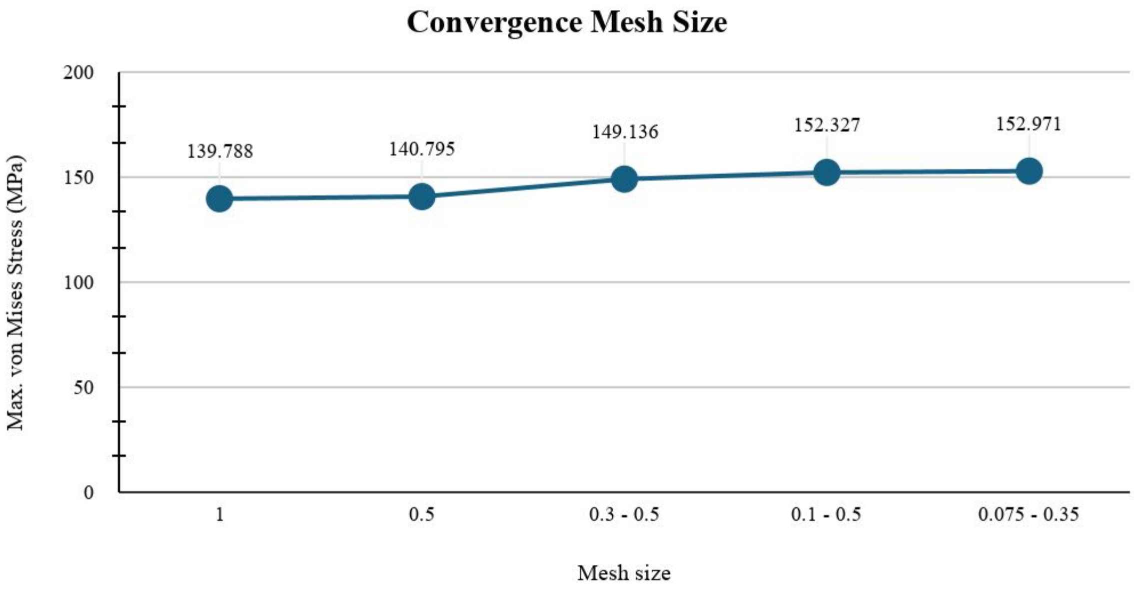


Global Mesh
Seed: 1mm

Convergence Maximum Stress

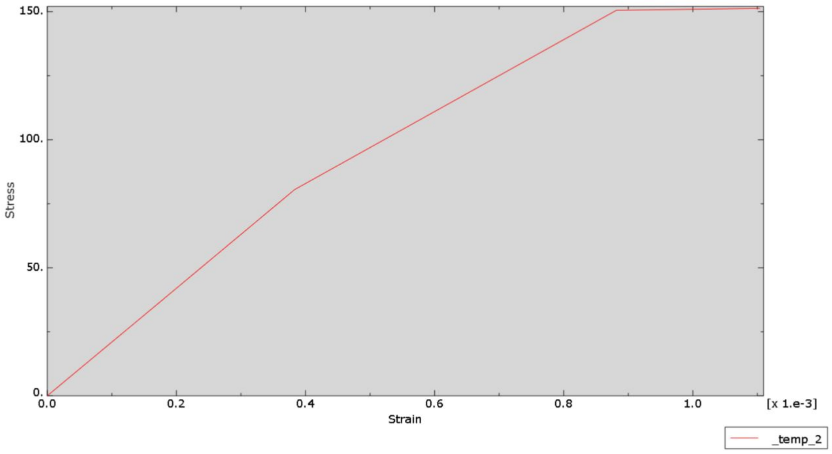
Steps	Global Mesh Seed (mm)	Max. Von Mises Stress	Precision (%)
1	1	139.788	None
2	0.5	140.795	7.956958
3	0.3 - 0.5	151.998	2.139657762
4	0.1 - 0.5	152.327	0.4227746887
5	0.075 - 0.35	152.971	None

Convergence Maximum Stress

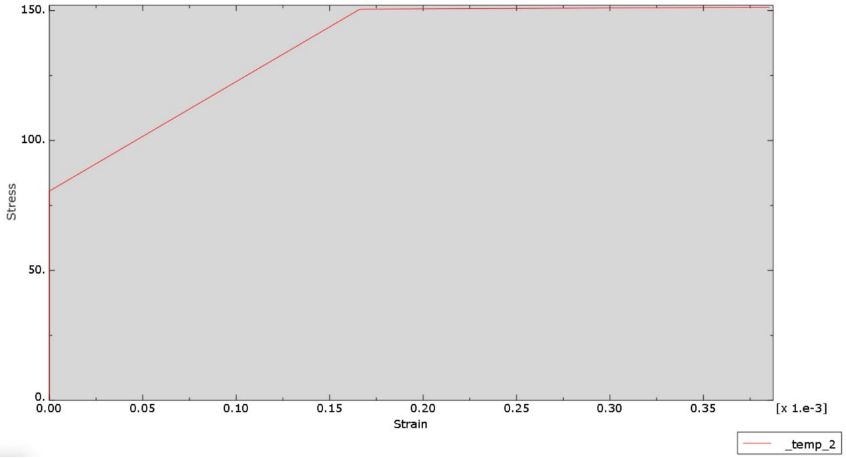


The mesh convergence study confirms that the solution is stable. The max von Mises stress stabilizes at **approximately 152 -155 MPa.** The low percentage change between the final mesh refinement steps **(mesh sizes 0.075 and 0.35)** validates that the result is independent of the discretization, ensuring the accuracy of the structural integrity assessment.

Stress and Strain



Yielding Stress



The primary difference lies in the **X-axis scaling and interpretation**.

- The left plot captures the entire deformation history including initial elastic behavior,
- whereas the right plot zooms into the plastic regime or represents a simplified material definition where elastic strains are ignored or shifted to evaluate pure yielding mechanics.

Maximum Weight

Load yield = Applied Load x (Yield Strength / Calculated Stress)

Load yield = 1000 x (250 / 152.971) = 1634.29 kg

Conclusion

Executive Conclusion: Structural Validation

- **Simulation Validation & Convergence:** To ensure the reliability of the numerical results, a systematic mesh convergence study was performed. The maximum von Mises stress asymptotically stabilized as the mesh was refined, with the relative error dropping to a negligible **0.095%** at the finest local element size of **0.35-0.0075 mm**. This mathematical convergence establishes high confidence that the final recorded peak stress of **152.971 MPa** is highly accurate and completely independent of mesh discretization.
- **Structural Integrity & Margin of Safety:** Under the standard design operating load of **1,000 kg**, the maximum induced stress (**152.93 MPa**) remains significantly below the material's structural yield strength of **250 MPa**. This indicates that the hook operates entirely within its linear elastic regime during standard deployment, providing a structural Factor of Safety (FoS) of **1.63**.
- **Ultimate Operational Yield Threshold:** Through linear scaling of the converged stress state, the ultimate operational load limit prior to the onset of plastic yielding was mathematically determined to be **1634.29 kg**.